

JEE MAIN REVISION MASTER: COMPLEX NUMBERS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 96 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Modulus, Argument & Polar/Euler Form	66 PYQs	Complex Basics
Very High	De Moivre's Theorem & Cube Roots of Unity	32 PYQs	Roots of Unity

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Modulus, Argument & Polar/Euler Form	$z = x + iy = r e^{i\theta}$, $ z ^2 = z \bar{z}$, $\text{Arg}(z_1 z_2) = \text{Arg}(z_1) + \text{Arg}(z_2)$, $ z_1 - z_2 \leq z_1 \pm z_2 \leq z_1 + z_2 $
De Moivre's Theorem & Cube Roots of Unity	$\omega^3 = 1$, $1 + \omega + \omega^2 = 0$, $\omega^2 = \bar{\omega}$, $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Modulus, Argument & Polar/Euler Form (Very High Yield — 66 PYQs)

Core Principle: Complex plane geometry, modulus properties, principal argument in $(-\pi, \pi]$, and triangle inequalities.

■ **Caution / JEE Pitfall:** Principal argument $\text{Arg}(z)$ must lie strictly in $(-\pi, \pi]$; if z is in the second quadrant, $\text{Arg}(z) = \pi - \tan^{-1}|y/x|$.

■ **Speed Trick:** Geometric interpretation: $|z - z_0| = r$ represents a circle centered at z_0 with radius r .

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "If $z = \frac{\sqrt{3} + i}{2}$, $i = \sqrt{-1}$, then $(z^{201} - i)^8$ is equal to..." [Ans: N/A]
- [2026 Evening Shift] Q#2 (MCQ): "Let $S = \{z \in \mathbb{C} : z^2 + \sqrt{6}iz - 3 = 0\}$. Then $\underset{z \in S}{\min} |z|^8$ is equal to :..." [Ans: N/A]
- [2026 Evening Shift] Q#2 (MCQ): "Let $S = \{z \in \mathbb{C} : z^2 + 4z + 16 = 0\}$. Then $\underset{z \in S}{\min} |z + \sqrt{3}i|^2$ is equal to :..." [Ans: N/A]

3.2 De Moivre's Theorem & Cube Roots of Unity (Very High Yield — 32 PYQs)

Core Principle: Algebraic and geometric properties of cube roots of unity forming an equilateral triangle on the unit circle, and De Moivre powers.

■ **Caution / JEE Pitfall:** $1 + \omega^n + \omega^{2n} = 3$ if n is a multiple of 3, and equals 0 if n is NOT a multiple of 3.

■ **Speed Trick:** Factorization identity: $x^3 - 1 = (x - 1)(x - \omega)(x - \omega^2)$, and $x^2 + x + 1 = (x - \omega)(x - \omega^2)$.

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (MCQ):** "The number of values of $z \in \mathbb{C}$, satisfying the equations $|z - (4 + 8i)| = \sqrt{10}$ and $|z - (3 + 5i)| + |z - (5 + 11i)| = 4\sqrt{5}$, is" [Ans: N/A]
- **[2026 Morning Shift] Q#1 (MCQ):** "Let $a, b \in \mathbb{C}$. Let α, β be the roots of the equation $x^2 + ax + b = 0$. If $\beta - \alpha = \sqrt{11}$ and $\beta^2 - \alpha^2 = 3i\sqrt{11}$..." [Ans: B]
- **[2026 Morning Shift] Q#1 (MCQ):** "Let x and y be real numbers such that $50((2x / 1 + 3i) - (y / 1 - 2i)) = 31 + 17i, i = \sqrt{-1}$. Then the value of $10(x - 3y)$ is" [Ans: D]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 5th April Morning Shift (Q#1)

Let $a, b \in \mathbb{C}$. Let α, β be the roots of the equation $x^2 + ax + b = 0$. If $\beta - \alpha = \sqrt{11}$ and $\beta^2 - \alpha^2 = 3i\sqrt{11}$, then $(\beta^3 - \alpha^3)^2$ is equal to:

- (A) 160
- (B) 176 [Correct]
- (C) 194
- (D) 187

Correct Answer: B

Solution Analysis: Given: α, β are roots of $x^2 + ax + b = 0$ $\beta - \alpha = \sqrt{11}$ $\beta^2 - \alpha^2 = 3i\sqrt{11}$ We have to find $(\beta^3 - \alpha^3)^2$. Now, $\beta^2 - \alpha^2 = (\beta - \alpha)(\beta + \alpha)$ Substituting the given values, $3i\sqrt{11} = \sqrt{11}(\beta + \alpha)$ Since $\sqrt{11} \neq 0$, we get $\beta + \alpha = 3i$ Now use the identity $\beta^3 - \alpha^3 = (\beta - \alpha)(\beta^2 + \alpha\beta + \alpha^2)$ So first let us find $\beta^2 + \alpha\beta + \alpha^2$. We know $(\beta + \alpha)^2 = \beta^2 + \alpha^2 + 2\alpha\beta$ Also, $\frac{\alpha + \beta}{\alpha\beta} = \frac{-(a)}{b}$

Benchmark Problem 2: JEE Main 2026 (Online) 4th April Morning Shift (Q#3)

Let z be a complex number such that $|z + 2| = |z - 2|$ and $\arg((z + 3 / z - i)) = (\pi / 4)$. Then $|z|^2$ is equal to:

- (A) 9 [Correct]
- (B) 4
- (C) 5
- (D) 1

Correct Answer: A

Solution Analysis: Let $z = x + iy$ where $x, y \in \mathbb{R}$. We are given: $|z + 2| = |z - 2|$ and $\arg((z + 3 / z - i)) = (\pi / 4)$ We will use these one by one. 1. Using $|z + 2| = |z - 2|$ Substitute $z = x + iy$: $|x + iy + 2| = |x + iy - 2|$ So, $\sqrt{(x+2)^2 + y^2} = \sqrt{(x-2)^2 + y^2}$ Squaring both sides: $(x+2)^2 + y^2 = (x-2)^2 + y^2$ $x^2 + 4x + 4 = x^2 - 4x + 4$ $8x = 0$ $x = 0$ Hence, $z = iy$ So the point lies on the imaginary axis. 2. Using the argument condition Now, $(z + 3 / z - i) = (3 + iy / i(y - 1))$ We are given that its argument is $(\pi / 4)$. A complex number has argument $(\pi / 4)$ when its real and imaginary parts are equal.